Ibon Gracia

PhD Student in Aerospace Engineering Sciences (Autonomous Systems)

ibon.gracia@colorado.edu | Boulder, Colorado, USA | Website | Google Scholar | LinkedIn

Research Interests

Safe and assured autonomy
Robotics and motion planning
Formal methods for control
Stochastic systems
Data-driven control and verification

Education

2023 – Present
Boulder, CO, USA

PhD in Aerospace Engineering Sciences
University of Colorado Boulder (CU Boulder)
Focus: Autonomous Systems
Advisor: Morteza Lahijanian
Expected graduation date: 05/2027

09/2020 – 10/2022
Delft, The Netherlands

MSc in Systems and Control
Delft University of Technology (TU Delft)
Thesis advisors: Dimitris Boskos, Luca Laurenti

09/2016 – 09/2019
Madrid, Spain

BSc in Aerospace Engineering
Technical University of Madrid (UPM)

09/2012 – 09/2016
Madrid, Spain

BSc in Mechanical Engineering
Technical University of Madrid (UPM)

Publications

Manuscripts Under Preparation

- **Gracia, I.**, Mazouz, R., & Lahijanian, M. Sound Motion Planning for Continuous-time Stochastic Systems under Digital Control. Targeting *IEEE Robotics and Automation Letters*
- **Gracia, I.**, & Lahijanian, M. Data-Driven UMDP Abstraction and Asymptotically Optimal Control Synthesis for Unknown Stochastic Systems. Targeting *ACM International Conference on Hybrid Systems: Computation and Control*
- Badings, T., **Gracia, I.**, Nazeri, M., Abate, A., & Lahijanian, M. Data-Driven IMDP and 2I-MDP Abstraction Tool. Targeting *ACM International Conference on Hybrid Systems: Computation and Control*

Manuscripts Under Review

- **Gracia, I.**, Ho, Q. H., Laurenti, L., & Lahijanian, M. (2026). Provably Safe Motion Planning Under Unknown Disturbances. Manuscript under review in *IEEE Transactions in Robotics*. arXiv preprint

Journal Publications

- **Gracia, I.**, & Lahijanian, M. (2025). Beyond Interval MDPs: Tight and Efficient Abstractions of Stochastic Systems. Provisionally accepted for publication in *Automatica*. arXiv preprint
- **Gracia, I.**, Boskos, D., Lahijanian, M., Laurenti, L., & Mazo Jr, M. (2025). Efficient strategy synthesis for switched stochastic systems with distributional uncertainty. *Nonlinear Analysis: Hybrid Systems*, 55, 101554 . arXiv version

Conference Publications

- **Gracia, I.**, & Lahijanian, M. (2026). On the Optimality of Uncertain MDP Abstractions. To appear in *IEEE Conference for Decision and Control*. arXiv preprint
- Kong, C. W., Escobar, S., **Gracia, I.**, McMahon, J., & Lahijanian, M. (2026). Training with hard constraints: Learning neural certificates and controllers for SDEs. In *International Conference on Neuro-symbolic Systems (NeuS). arXiv version*
- **Gracia, I.**, & Lahijanian, M. (2025). Data-Driven Control via Conditional Mean Embeddings: Formal Guarantees via Uncertain MDP Abstraction. In *ACM International Conference on Hybrid Systems: Computation and Control. arXiv version* .
- **Gracia, I.**, Boskos, D., Laurenti, L., & Mazo Jr, M. (2023, May). Distributionally robust strategy synthesis for switched stochastic systems. In *Proceedings of the 26th ACM International Conference on Hybrid Systems: Computation and Control* (pp. 1-10). **Best paper award**
- **Gracia, I.**, Boskos, D., Laurenti, L., & Lahijanian, M. (2024, June). Data-driven strategy synthesis for stochastic systems with unknown nonlinear disturbances. In *6th Annual Learning for Dynamics & Control Conference* (pp. 1633-1645). *PMLR* . arXiv version
- **Gracia, I.**, Laurenti, L., Mazo Jr., M., Abate, A. and Lahijanian, M. (2025). Temporal Logic Control for Nonlinear Stochastic Systems Under Unknown Disturbances. *Proceedings of the 7th Annual Learning for Dynamics & Control Conference (L4DC). PMLR . arXiv version*

Awards

2023 — **Best Paper Award**
ACM International Conference on Hybrid Systems: Computation and Control

Teaching Experience

08/2026 – Present, Boulder, CO, USA — **ASEN 5044: Statistical Estimation for Dynamical Systems**
University of Colorado Boulder
Teaching Assistant

05/2026 – 07/2026, University of Colorado Boulder — **ASEN 3728 — Aircraft Dynamics, University of Colorado Boulder**
Summer 2026
Instructor of record

04/2026 – 04/2026, Boulder, CO, USA — **ASEN 6519 — Safe Autonomy Amid Uncertainty**
University of Colorado Boulder
Guest lecturer

01/2026 – 05/2026, Boulder, CO, USA — **ASEN 3728: Aircraft Dynamics**
University of Colorado Boulder
Teaching Assistant

08/2025 – 12/2025, Boulder, CO, USA — **ASEN 3711: Aerodynamics**
University of Colorado Boulder
Teaching Assistant

07/2021 – 09/2021, Delft, The Netherlands — **SC42155: Modelling of Dynamical Systems**
Delft University of Technology (TU Delft)
Student Assistant
Helped the professors to extend the reader of a course in system modeling with additional theory and exercises.

Industry Experience

03/2019 – 08/2020
Madrid, Spain

Capgemini DEMS
Electronic Flight Control Systems Engineer
R&D project on the automation of an an air-to-air refueling system. Duties involving flight control systems specification, modeling, verification and validation, hardware-in-the-loop testing, as well as support to departments such as Software, Flight Dynamics and Flight Test.

Conference Presentations

- "Data-Driven Control via Conditional Mean Embeddings: Formal Guarantees via Uncertain MDP Abstraction," *ACM HSCC* 2026
- "Training with Hard Constraints: Learning Neural Certificates and Controllers for SDEs," *NeuS*, 2026
- "Temporal Logic Control for Nonlinear Stochastic Systems Under Unknown Disturbances," *L4DC*, 2025. **Poster**
- ""Data-driven strategy synthesis for stochastic systems with unknown nonlinear disturbances", *L4DC* 2024. **Poster**
- "Distributionally robust strategy synthesis for switched stochastic systems", *ACM HSCC* 2023

Professional Service

Reviewer for Automatica, Hybrid Systems: Computation and Control (HSCC), Conference on Decision and Control (CDC), Learning for Dynamics and Control (L4DC), ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS), American Control Conference (ACC), European Control Conference (ECC), IEEE International Conference on Robotics and Automation (ICRA)

Programming Skills

- Matlab/Simulink
- C++
- LaTeX
- Python
- Julia

Languages

English

Spanish